r_e

CHAPTER

7 BJT Transistor Modeling

7.1 INTRODUCTION

The basic construction, appearance, and characteristics of the transistor were introduced in Chapter 3. The dc biasing of the device was then examined in detail in Chapter 4. We now begin to examine the *small-signal* ac response of the BJT amplifier by reviewing the *models* most frequently used to represent the transistor in the sinusoidal ac domain.

One of our first concerns in the sinusoidal ac analysis of transistor networks is the magnitude of the input signal. It will determine whether *small-signal* or *large-signal* techniques should be applied. There is no set dividing line between the two, but the application—and the magnitude of the variables of interest relative to the scales of the device characteristics—will usually make it quite clear which method is appropriate. The small-signal technique is introduced in this chapter, and large-signal applications are examined in Chapter 16.

There are two models commonly used in the small-signal ac analysis of transistor networks: the r_e model and the *hybrid equivalent* model. This chapter not only introduces both models but defines the role of each and the relationship between the two.

7.2 AMPLIFICATION IN THE AC DOMAIN

It was demonstrated in Chapter 3 that the transistor can be employed as an amplifying device. That is, the output sinusoidal signal is greater than the input signal or, stated another way, the output ac power is greater than the input ac power. The question then arises as to how the ac power output can be greater than the input ac power? Conservation of energy dictates that over time the total power output, P_o, of a system cannot be greater than its power input, P_i, and that the efficiency defined by $\eta = P_o/P_i$ cannot be greater than 1. The factor missing from the discussion above that permits an ac power output greater than the input ac power is the applied dc power. It is a contributor to the total output power even though part of it is dissipated by the device and resistive elements. In other words, there is an "exchange" of dc power to the ac domain that permits establishing a higher output ac power. In fact, a *conversion efficiency* is defined by $\eta = P_{o(\text{ac})}/P_{i(\text{dc})}$, where $P_{o(\text{ac})}$ is the ac power to the load and $P_{i(\text{dc})}$ is the dc power supplied.

Perhaps the role of the dc supply can best be described by first considering the simple dc network of Fig. 7.1. The resulting direction of flow is indicated in the figure with a plot of the current i versus time. Let us now insert a control mechanism

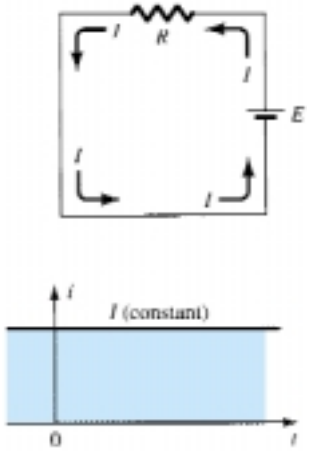


Figure 7.1 Steady current established by a dc supply.

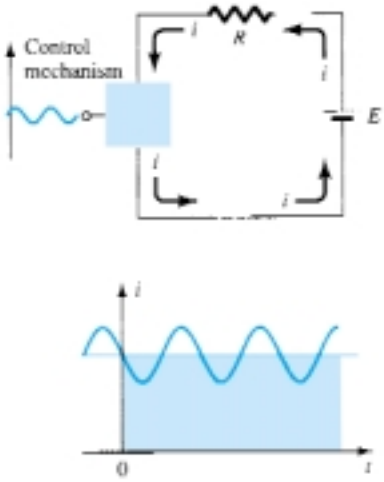


Figure 7.2 Effect of a control element on the steady-state flow of the electrical system of Fig. 7.1.

such as that shown in Fig. 7.2. The control mechanism is such that the application of a relatively small signal to the control mechanism can result in a much larger oscillation in the output circuit. For the system of Fig. 7.2, the peak value of the oscillation is controlled by the established dc level. Any attempt to exceed the limit set by the dc level will result in a "clipping" (flattening) of the peak region of the output signal. In total, therefore, proper amplifier design requires that the dc and ac components be sensitive to each other's requirements and limitations.

However, it is indeed fortunate that transistor small-signal amplifiers can be considered linear for most applications, permitting the use of the superposition theorem to isolate the dc analysis from the ac analysis.

7.3 BJT TRANSISTOR MODELING

The key to transistor small-signal analysis is the use of equivalent circuits (models) to be introduced in this chapter.

A model is the combination of circuit elements, properly chosen, that best approximates the actual behavior of a semiconductor device under specific operating conditions.

Once the ac equivalent circuit has been determined, the graphical symbol of the device can be replaced in the schematic by this circuit and the basic methods of ac circuit analysis (mesh analysis, nodal analysis, and Thévenin's theorem) can be applied to determine the response of the circuit.

There are two schools of thought in prominence today regarding the equivalent circuit to be substituted for the transistor. For many years the industrial and educational institutions relied heavily on the *hybrid parameters* (to be introduced shortly). The hybrid-parameter equivalent circuit continues to be very popular, although it must now share the spotlight with an equivalent circuit derived directly from the operating conditions of the transistor—the r_e model. Manufacturers continue to specify the hybrid parameters for a particular operating region on their specification sheets. The parameters (or components) of the r_e model can be derived directly from the hybrid parameters in this region. However, the hybrid equivalent circuit suffers from being limited to a particular set of operating conditions if it is to be considered accurate. The parameters of the other equivalent circuit can be determined for any region of operation within the active region and are not limited by the single set of parameters provided by the specification sheet. In turn, however, the r_e model fails to account for the output impedance level of the device and the feedback effect from output to input.

Since both models are used extensively today, they are both examined in detail in this text. In some analysis and examples the hybrid model will be employed, while in others the r_e model will be used exclusively. The text will make every effort, however, to show how closely related the two models are and how a proficiency with one leads to a natural proficiency with the other.

In an effort to demonstrate the effect that the ac equivalent circuit will have on the analysis to follow, consider the circuit of Fig. 7.3. Let us assume for the moment that the small-signal ac equivalent circuit for the transistor has already been determined. Since we are interested only in the ac response of the circuit, all the dc supplies can be replaced by a zero-potential equivalent (short circuit) since they determine only the dc (quiescent level) of the output voltage and not the magnitude of the swing of the ac output. This is clearly demonstrated by Fig. 7.4. The dc levels were simply important for determining the proper *Q*-point of operation. Once determined, the dc levels can be ignored in the ac analysis of the network. In addition, the coupling capacitors C_1 and C_2 and bypass capacitor C_3 were chosen to have a very small

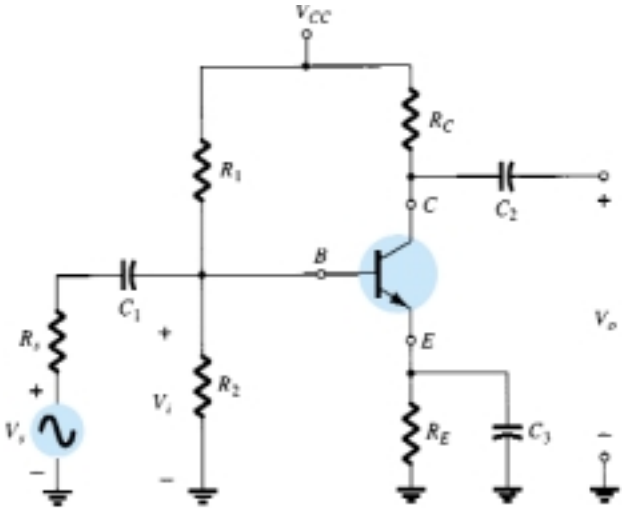


Figure 7.3 Transistor circuit under examination in this introductory discussion.

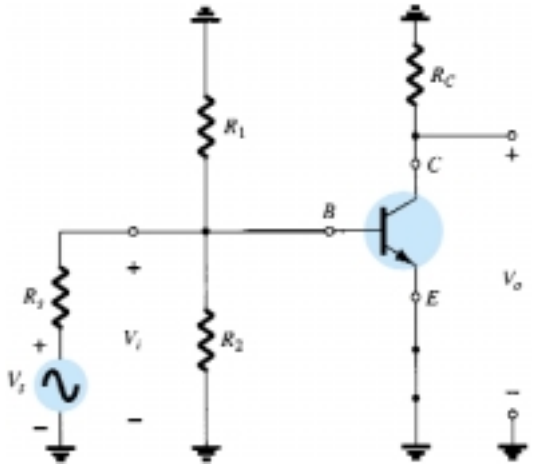


Figure 7.4 The network of Fig. 7.3 following removal of the dc supply and insertion of the short-circuit equivalent for the capacitors.

reactance at the frequency of application. Therefore, they too may for all practical purposes be replaced by a low-resistance path or a short circuit. Note that this will result in the "shorting out" of the dc biasing resistor R_E. Recall that capacitors assume an "open-circuit" equivalent under dc steady-state conditions, permitting an isolation between stages for the dc levels and quiescent conditions.

If we establish a common ground and rearrange the elements of Fig. 7.4, R_1 and R_2 will be in parallel and R_C will appear from collector to emitter as shown in Fig. 7.5. Since the components of the transistor equivalent circuit appearing in Fig. 7.5 employ familiar components such as resistors and independent controlled sources,

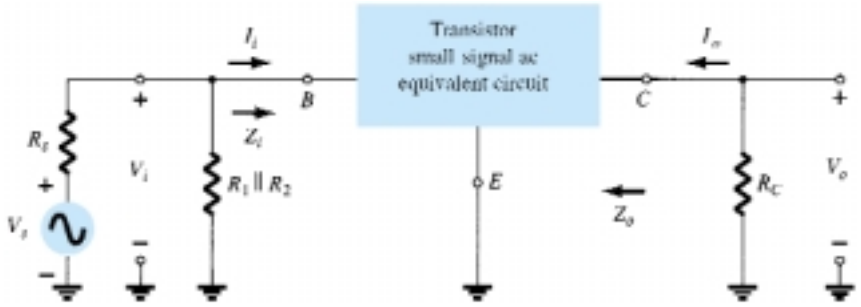


Figure 7.5 Circuit of Fig. 7.4 redrawn for small-signal ac analysis.

analysis techniques such as superposition, Thévenin's theorem, and so on, can be applied to determine the desired quantities.

Let us further examine Fig. 7.5 and identify the important quantities to be determined for the system. Since we know that the transistor is an amplifying device, we would expect some indication of how the output voltage V_o is related to the input voltage V_i—the *voltage gain*. Note in Fig. 7.5 for this configuration that $I_i = I_b$ and $I_o = I_c$, which define the *current gain* $A_i = I_o/I_i$. The input impedance Z_i and output impedance Z_o will prove particularly important in the analysis to follow. A great deal more will be offered about these parameters in the sections to follow.

In summary, therefore, the ac equivalent of a network is obtained by:

1. *Setting all dc sources to zero and replacing them by a short-circuit equivalent*
2. *Replacing all capacitors by a short-circuit equivalent*
3. *Removing all elements bypassed by the short-circuit equivalents introduced by steps 1 and 2*
4. *Redrawing the network in a more convenient and logical form*

In the sections to follow, the r_e and hybrid equivalent circuits will be introduced to complete the ac analysis of the network of Fig. 7.5.

7.4 THE IMPORTANT PARAMETERS: Z_i, Z_o, A_v, A_i

Before investigating the equivalent circuits for BJTs in some detail, let us concentrate on those parameters of a two-port system that are of paramount importance from an analysis and design viewpoint. For the two-port (two pairs of terminals) system of Fig. 7.6, the input side (the side to which the signal is normally applied) is to the left and the output side (where the load is connected) is to the right. In fact, for most electrical and electronic systems, the general flow is usually from the left to the right. For both sets of terminals, the impedance between each pair of terminals under normal operating conditions is quite important.

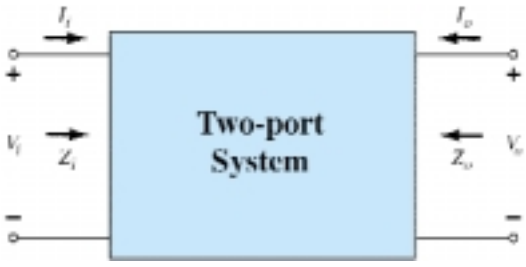


Figure 7.6 Two-port system.

Input Impedance, Z_i

For the input side, the input impedance Z_i is defined by Ohm's law as the following:

$$Z_i = \frac{V_i}{I_i} \tag{7.1}$$

If the input signal V_i is changed, the current I_i can be computed using the same level of input impedance. In other words:

For small-signal analysis, once the input impedance has been determined the same numerical value can be used for changing levels of applied signal.

In fact, we will find in the sections to follow that the input impedance of a transistor can be approximately determined by the dc biasing conditions—conditions that do not change simply because the magnitude of the applied ac signal has changed.

It is particularly noteworthy that for frequencies in the low to mid-range (typically $\leq$100 kHz):

The input impedance of a BJT transistor amplifier is purely resistive in nature and, depending on the manner in which the transistor is employed, can vary from a few ohms to megohms.

In addition:

An ohmmeter cannot be used to measure the small-signal ac input impedance since the ohmmeter operates in the dc mode.

Equation (7.1) is particularly useful in that it provides a method for measuring the input resistance in the ac domain. For instance, in Fig. 7.7 a sensing resistor has been added to the input side to permit a determination of I_i using Ohm's law. An oscilloscope or sensitive digital multimeter (DMM) can be used to measure the voltage V_s and V_i. Both voltages can be the peak-to-peak, peak, or rms values, as long as both levels use the same standard. The input impedance is then determined in the following manner:

$$I_i = \frac{V_s - V_i}{R_{\text{sense}}} \tag{7.2}$$

and

$$Z_i = \frac{V_i}{I_i} \tag{7.3}$$

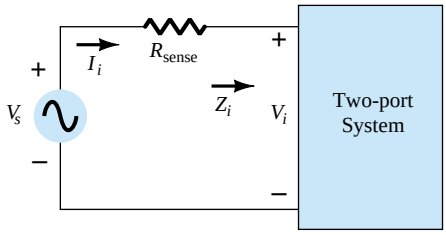


Figure 7.7 Determining Z_i.

The importance of the input impedance of a system can best be demonstrated by the network of Fig. 7.8. The signal source has an internal resistance of 600 Ω, and the system (possibly a transistor amplifier) has an input resistance of 1.2 kΩ. If the source were ideal ($R_s = 0$ Ω), the full 10 mV would be applied to the system, but

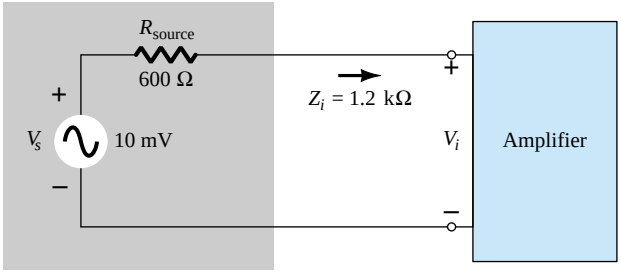


Figure 7.8 Demonstrating the impact of Z_i on an amplifier's response.

with a source impedance, the input voltage must be determined using the voltage-divider rule as follows:

$$V_i = \frac{Z_i V_s}{Z_i + R_{\text{source}}} = \frac{(1.2\ \text{k}\Omega)(10\ \text{mV})}{1.2\ \text{k}\Omega + 0.6\ \text{k}\Omega} = 6.67\ \text{mV}$$

Thus, only 66.7% of the full-input signal is available at the input. If Z_i were only 600 Ω, then $V_i = \frac{1}{2}(10\ \text{mV}) = 5\ \text{mV}$ or 50% of the available signal. Of course, if $Z_i = 8.2\ \text{k}\Omega$, V_i will be 93.2% of the applied signal. The level of input impedance, therefore, can have a significant impact on the level of signal that reaches the system (or amplifier). In the sections and chapters to follow, it will be demonstrated that the ac input resistance is dependent on whether the transistor is in the common-base, common-emitter, or common-collector configuration and on the placement of the resistive elements.

EXAMPLE 7.1 For the system of Fig. 7.9, determine the level of input impedance.

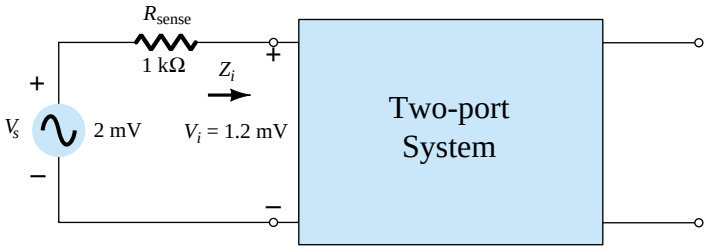


Figure 7.9 Example 7.1

Solution

$$I_i = \frac{V_s - V_i}{R_{\text{sense}}} = \frac{2\ \text{mV} - 1.2\ \text{mV}}{1\ \text{k}\Omega} = \frac{0.8\ \text{mV}}{1\ \text{k}\Omega} = 0.8\ \mu\text{A}$$

and

$$Z_i = \frac{V_i}{I_i} = \frac{1.2\ \text{mV}}{0.8\ \mu\text{A}} = \mathbf{1.5\ k\Omega}$$

Output Impedance, Z_o

The output impedance is naturally defined at the output set of terminals, but the manner in which it is defined is quite different from that of the input impedance. That is:

The output impedance is determined at the output terminals looking back into the system with the applied signal set to zero.

In Fig. 7.10, for example, the applied signal has been set to zero volts. To determine Z_o, a signal, V_s, is applied to the output terminals and the level of V_o is measured with an oscilloscope or sensitive DMM. The output impedance is then determined in the following manner:

$$I_o = \frac{V - V_o}{R_{\text{sense}}} \tag{7.4}$$

and

$$Z_o = \frac{V_o}{I_o} \tag{7.5}$$

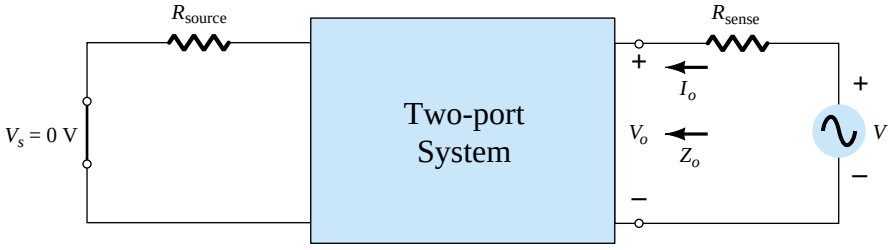


Figure 7.10 Determining Z_o.

In particular, for frequencies in the low to mid-range (typically ≤ 100 kHz):

The output impedance of a BJT transistor amplifier is resistive in nature and, depending on the configuration and the placement of the resistive elements, Z_o, can vary from a few ohms to a level that can exceed 2 MΩ.

In addition:

An ohmmeter cannot be used to measure the small-signal ac output impedance since the ohmmeter operates in the dc mode.

For amplifier configurations where significant gain in current is desired, the level of Z_o should be as large as possible. As demonstrated by Fig. 7.11, if $Z_o \gg R_L$, the majority of the amplifier output current will pass on to the load. It will be demonstrated in the sections and chapters to follow that Z_o is frequently so large compared to R_L that it can be replaced by an open-circuit equivalent.

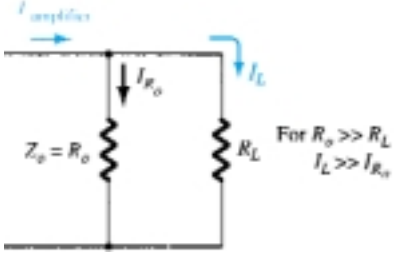


Figure 7.11 Effect of $Z_o = R_o$ on the load or output current I_L.

EXAMPLE 7.2

For the system of Fig. 7.12, determine the level of output impedance.

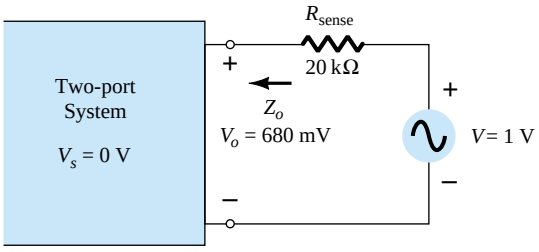


Figure 7.12 Example 7.2.

Solution

$$I_o = \frac{V - V_o}{R_{sense}} = \frac{1\text{ V} - 680\text{ mV}}{20\text{ k}\Omega} = \frac{320\text{ mV}}{20\text{ k}\Omega} = 16\ \mu\text{A}$$

and

$$Z_o = \frac{V_o}{I_o} = \frac{680\text{ mV}}{16\ \mu\text{A}} = \mathbf{42.5\ k\Omega}$$

Voltage Gain, A_v

One of the most important characteristics of an amplifier is the small-signal ac voltage gain as determined by

$$A_v = \frac{V_o}{V_i} \tag{7.6}$$

For the system of Fig. 7.13, a load has not been connected to the output terminals and the level of gain determined by Eq. (7.6) is referred to as the no-load voltage gain. That is,

$$A_{v_{NL}} = \left.\frac{V_o}{V_i}\right|_{R_L = \infty\ \Omega\ \text{(open circuit)}} \tag{7.7}$$

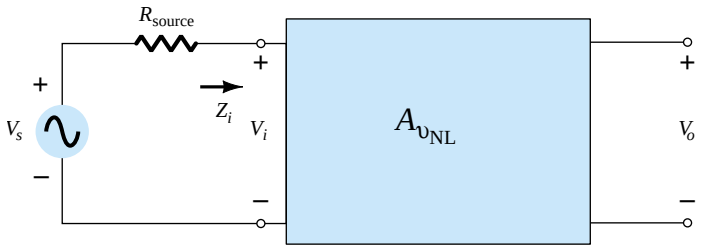


Figure 7.13 Determining the no-load voltage gain.

In Chapter 9 it will be demonstrated that:

For transistor amplifiers, the no-load voltage gain is greater than the loaded voltage gain.

For the system of Fig. 7.13 having a source resistance R_s, the level of V_i would first have to be determined using the voltage-divider rule before the gain V_o/V_s could be calculated. That is,

$$V_i = \frac{Z_i V_s}{Z_i + R_s}$$

with

$$\frac{V_i}{V_s} = \frac{Z_i}{Z_i + R_s}$$

and

$$A_{v_s} = \frac{V_o}{V_s} = \frac{V_i}{V_s} \cdot \frac{V_o}{V_i}$$

so that

$$A_{v_s} = \frac{V_o}{V_s} = \frac{Z_i}{Z_i + R_s} A_{v_{NL}} \tag{7.8}$$

Experimentally, the voltage gain A_{v_s} or $A_{v_{NL}}$ can be determined simply by measuring the appropriate voltage levels with an oscilloscope or sensitive DMM and substituting into the appropriate equation.

Depending on the configuration, the magnitude of the voltage gain for a loaded single-stage transistor amplifier typically ranges from just less than 1 to a few hundred. A multistage (multiunit) system, however, can have a voltage gain in the thousands.

EXAMPLE 7.3

For the BJT amplifier of Fig. 7.14, determine:
(a) V_i.
(b) I_i.
(c) Z_i.
(d) A_{v_s}.

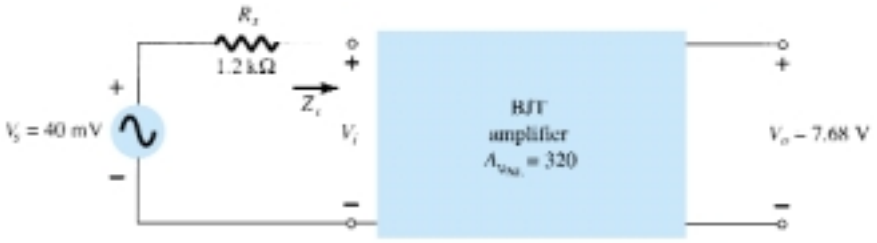


Figure 7.14 Example 7.3.

Solution

(a) $A_{v_{NL}} = \dfrac{V_o}{V_i}$ and $V_i = \dfrac{V_o}{A_{v_{NL}}} = \dfrac{7.68\text{ V}}{320} = \mathbf{24\text{ mV}}$

(b) $I_i = \dfrac{V_s - V_i}{R_s} = \dfrac{40\text{ mV} - 24\text{ mV}}{1.2\text{ k}\Omega} = \mathbf{13.33\ \mu A}$

(c) $Z_i = \dfrac{V_i}{I_i} = \dfrac{24\text{ mV}}{13.33\ \mu\text{A}} = \mathbf{1.8\ k\Omega}$

(d) $A_{v_s} = \dfrac{Z_i}{Z_i + R_s} A_{v_{NL}}$

$$= \frac{1.8\text{ k}\Omega}{1.8\text{ k}\Omega + 1.2\text{ k}\Omega}(320)$$

$$= \mathbf{192}$$

Current Gain, A_i

The last numerical characteristic to be discussed is the current gain defined by

$$A_i = \frac{I_o}{I_i} \tag{7.9}$$

Although typically the recipient of less attention than the voltage gain, it is, however, an important quantity that can have significant impact on the overall effectiveness of a design. In general:

For BJT amplifiers, the current gain typically ranges from a level just less than 1 to a level that may exceed 100.

For the loaded situation of Fig. 7.15,

$$I_i = \frac{V_i}{Z_i} \quad \text{and} \quad I_o = -\frac{V_o}{R_L}$$

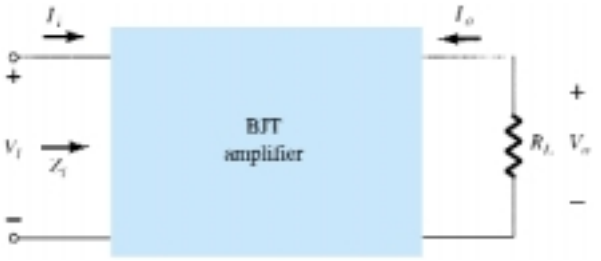


Figure 7.15 Determining the loaded current gain.

with $$A_i = \frac{I_o}{I_i} = -\frac{V_o/R_L}{V_i/Z_i} = -\frac{V_o Z_i}{V_i R_L}$$

and $$\boxed{A_i = -A_v \frac{Z_i}{R_L}} \qquad (7.10)$$

Eq. (7.10) allows the determination of the current gain from the voltage gain and the impedance levels.

Phase Relationship

The phase relationship between input and output sinusoidal signals is important for a variety of practical reasons. Fortunately, however:

For the typical transistor amplifier at frequencies that permit ignoring the effects of the reactive elements, the input and output signals are either 180° out of phase or in phase.

The reason for the either–or situation will become quite clear in the chapters to follow.

Summary

The parameters of primary importance for an amplifier have now been introduced: the input impedance Z_i, the output impedance Z_o, the voltage gain A_v, the current gain A_i, and the resulting phase relationship. Other factors, such as the applied frequency at the low and high ends of the frequency spectrum, will affect some of these parameters, but this will be discussed in Chapter 11. In the sections and chapters to follow, all the parameters will be determined for a variety of transistor networks to permit a comparison of the strengths and weaknesses for each configuration.

7.5 THE r_e TRANSISTOR MODEL

The r_e model employs a diode and controlled current source to duplicate the behavior of a transistor in the region of interest. Recall that a current-controlled current source is one where the parameters of the current source are controlled by a current elsewhere in the network. In fact, in general:

BJT transistor amplifiers are referred to as current-controlled devices.

Common Base Configuration

In Fig. 7.16a, a common-base *pnp* transistor has been inserted within the two-port structure employed in our discussion of the last few sections. In Fig. 7.16b, the r_e model for the transistor has been placed between the same four terminals. As noted in Section 7.3, the model (equivalent circuit) is chosen in such a way as to approximate the behavior of the device it is replacing in the operating region of interest. In other words, the results obtained with the model in place should be relatively close to those obtained with the actual transistor. You will recall from Chapter 3 that one junction of an operating transistor is forward-biased while the other is reverse-biased. The forward-biased junction will behave much like a diode (ignoring the effects of changing levels of V_{CE}) as verified by the curves of Fig. 3.7. For the base-to-emitter junction of the transistor of Fig. 7.16a, the diode equivalence of Fig. 7.16b between the same two terminals seems to be quite appropriate. For the output side, recall that the horizontal curves of Fig. 3.8 revealed that $I_c \cong I_e$ (as derived from $I_c = \alpha I_e$) for the range of values of V_{CE}. The current source of Fig. 7.16b establishes the fact that

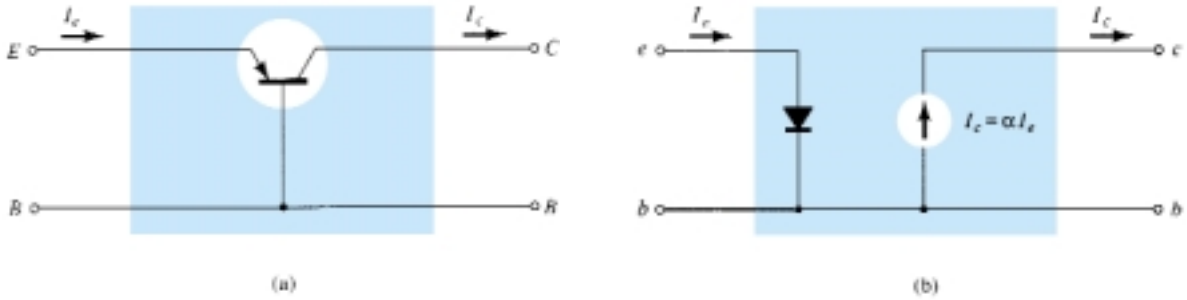


Figure 7.16 (a) Common-base BJT transistor; (b) r_e model for the configuration of Fig. 7.16a.

$I_c = \alpha I_e$, with the controlling current I_e appearing in the input side of the equivalent circuit as dictated by Fig. 7.16a. We have therefore established an equivalence at the input and output terminals with the current-controlled source, providing a link between the two—an initial review would suggest that the model of Fig. 7.16b is a valid model of the actual device.

Recall from Chapter 1 that the ac resistance of a diode can be determined by the equation $r_{ac} = 26\ \text{mV}/I_D$, where I_D is the dc current through the diode at the Q (quiescent) point. This same equation can be used to find the ac resistance of the diode of Fig. 7.16b if we simply substitute the emitter current as follows:

$$r_e = \frac{26\ \text{mV}}{I_E} \tag{7.11}$$

The subscript e of r_e was chosen to emphasize that it is the dc level of emitter current that determines the ac level of the resistance of the diode of Fig. 7.16b. Substituting the resulting value of r_e in Fig. 7.16b will result in the very useful model of Fig. 7.17.

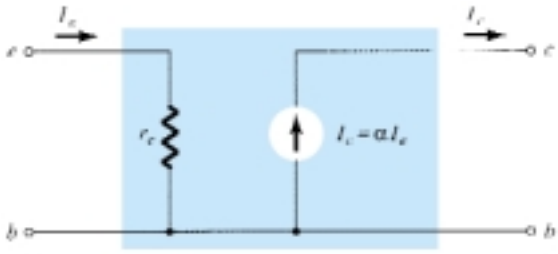


Figure 7.17 Common-base r_e equivalent circuit.

Due to the isolation that exists between input and output circuits of Fig. 7.17, it should be fairly obvious that the input impedance Z_i for the common-base configuration of a transistor is simply r_e. That is,

$$Z_i = r_e \quad (CB) \tag{7.12}$$

For the common-base configuration, typical values of Z_i range from a few ohms to a maximum of about 50 Ω.

For the output impedance, if we set the signal to zero, then $I_e = 0$ A and $I_c = \alpha I_e = \alpha(0\ \text{A}) = 0$ A, resulting in an open-circuit equivalence at the output terminals. The result is that for the model of Fig. 7.17,

$$Z_o \cong \infty\ \Omega \quad (CB) \tag{7.13}$$

In actuality:

For the common-base configuration, typical values of Z_o are in the megohm range.

The output resistance of the common-base configuration is determined by the slope of the characteristic lines of the output characteristics as shown in Fig. 7.18. Assuming the lines to be perfectly horizontal (an excellent approximation) would result in the conclusion of Eq. (7.13). If care were taken to measure Z_o graphically or experimentally, levels typically in the range 1- to 2-MΩ would be obtained.

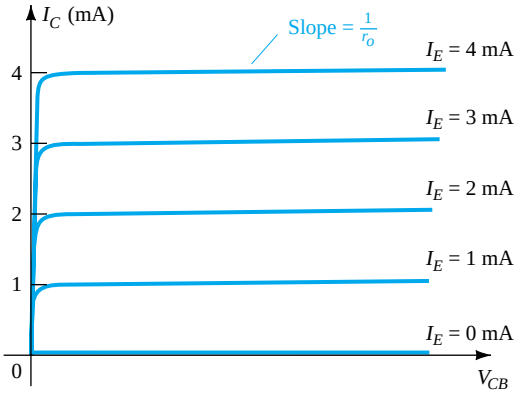


Figure 7.18 Defining Z_o.

In general, for the common-base configuration the input impedance is relatively small and the output impedance quite high.

The voltage gain will now be determined for the network of Fig. 7.19.

$$V_o = -I_o R_L = -(-I_c)R_L = \alpha I_e R_L$$

and

$$V_i = I_e Z_i = I_e r_e$$

so that

$$A_v = \frac{V_o}{V_i} = \frac{\alpha I_e R_L}{I_e r_e}$$

and

$$\boxed{A_v = \frac{\alpha R_L}{r_e} \cong \frac{R_L}{r_e}}_{CB} \tag{7.14}$$

For the current gain,

$$A_i = \frac{I_o}{I_i} = \frac{-I_c}{I_e} = -\frac{\alpha I_e}{I_e}$$

and

$$\boxed{A_i = -\alpha \cong -1}_{CB} \tag{7.15}$$

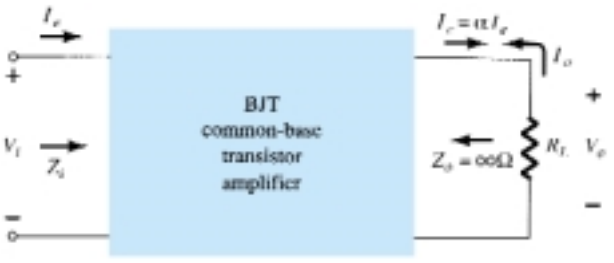


Figure 7.19 Defining $A_v = V_o/V_i$ for the common-base configuration.

The fact that the polarity of the voltage V_o as determined by the current I_c is the same as defined by Fig. 7.19 (i.e., the negative side is at ground potential) reveals that V_o and V_i are *in phase* for the common-base configuration. For an *npn* transistor in the common-base configuration, the equivalence would appear as shown in Fig. 7.20.

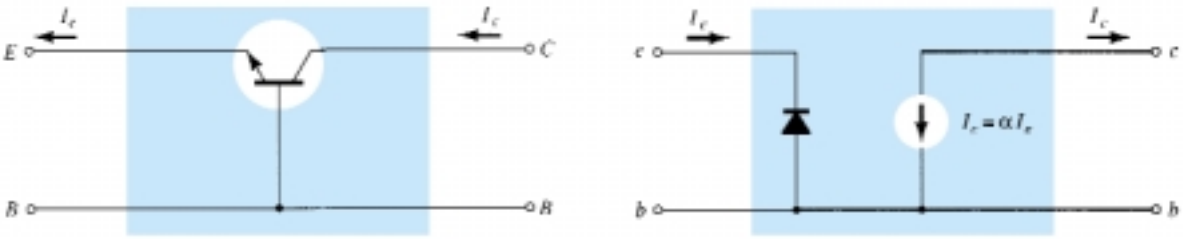


Figure 7.20 Approximate model for a common-base *npn* transistor configuration.

EXAMPLE 7.4

For a common-base configuration of Fig. 7.17 with $I_E = 4$ mA, $\alpha = 0.98$, and an ac signal of 2 mV applied between the base and emitter terminals:

(a) Determine the input impedance.
(b) Calculate the voltage gain if a load of 0.56 kΩ is connected to the output terminals.
(c) Find the output impedance and current gain.

Solution

(a) $r_e = \dfrac{26 \text{ mV}}{I_E} = \dfrac{26 \text{ mV}}{4 \text{ mA}} = \mathbf{6.5\ \Omega}$

(b) $I_i = I_e = \dfrac{V_i}{Z_i} = \dfrac{2 \text{ mV}}{6.5\ \Omega} = 307.69\ \mu\text{A}$

$V_o = I_c R_L = \alpha I_e R_L = (0.98)(307.69\ \mu\text{A})(0.56\ \text{k}\Omega)$
$= 168.86 \text{ mV}$

and $A_v = \dfrac{V_o}{V_i} = \dfrac{168.86 \text{ mV}}{2 \text{ mV}} = \mathbf{84.43}$

or from Eq. (7.14),

$$A_v = \frac{\alpha R_L}{r_e} = \frac{(0.98)(0.56\ \text{k}\Omega)}{6.5\ \Omega} = \mathbf{84.43}$$

(c) $Z_o \cong \infty\ \Omega$

$A_i = \dfrac{I_o}{I_i} = -\alpha = \mathbf{-0.98}$ as defined by Eq. (7.15)

Common Emitter Configuration

For the common-emitter configuration of Fig. 7.21a, the input terminals are the base and emitter terminals, but the output set is now the collector and emitter terminals. In addition, the emitter terminal is now common between the input and output ports of the amplifier. Substituting the r_e equivalent circuit for the *npn* transistor will result in the configuration of Fig. 7.21b. Note that the controlled-current source is still connected between the collector and base terminals and the diode between the base and

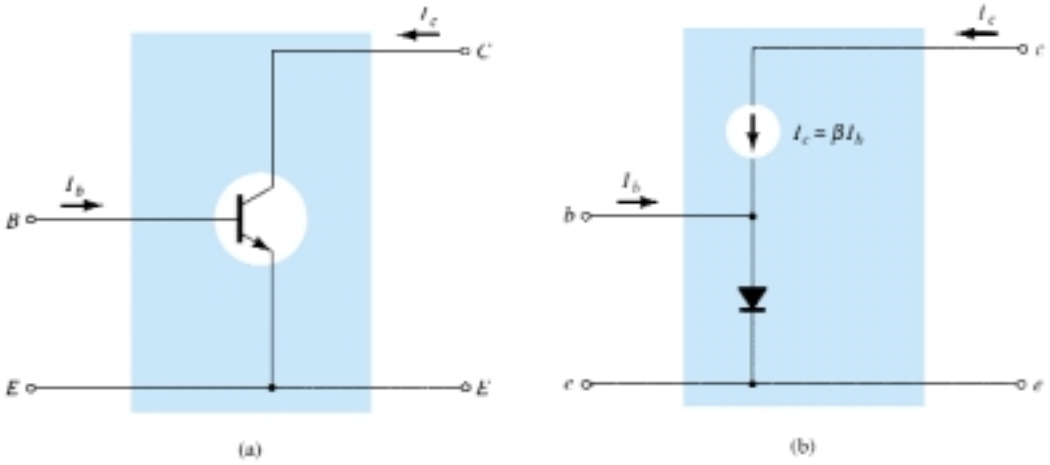


Figure 7.21 (a) Common-emitter BJT transistor; (b) approximate model for the configuration of Fig. 7.21a.

emitter terminals. In this configuration, the base current is the input current while the output current is still I_c. Recall from Chapter 3 that the base and collector currents are related by the following equation:

$$\boxed{I_c = \beta I_b} \tag{7.16}$$

The current through the diode is therefore determined by

$$I_e = I_c + I_b = \beta I_b + I_b$$

and

$$\boxed{I_e = (\beta + 1)I_b} \tag{7.17}$$

However, since the ac beta is typically much greater than 1, we will use the following approximation for the current analysis:

$$\boxed{I_e \cong \beta I_b} \tag{7.18}$$

The input impedance is determined by the following ratio:

$$Z_i = \frac{V_i}{I_i} = \frac{V_{be}}{I_b}$$

The voltage V_{be} is across the diode resistance as shown in Fig. 7.22. The level of r_e is still determined by the dc current I_E. Using Ohm's law gives

$$V_i = V_{be} = I_e r_e \cong \beta I_b r_e$$

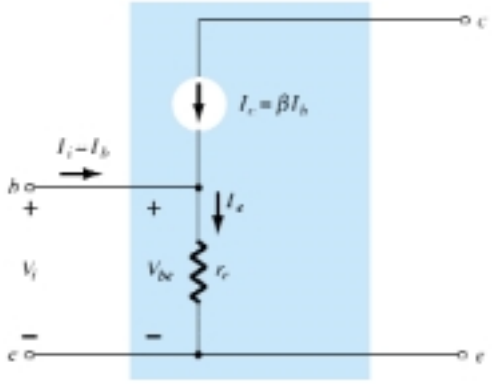


Figure 7.22 Determining Z_i using the approximate model.

Substituting yields

$$Z_i = \frac{V_{be}}{I_b} \cong \frac{\beta I_b r_e}{I_b}$$

and

$$\boxed{Z_i \cong \beta r_e}_{CE} \tag{7.19}$$

In essence, Eq. (7.19) states that the input impedance for a situation such as shown in Fig. 7.23 is beta times the value of r_e. In other words, a resistive element in the emitter leg is reflected into the input circuit by a multiplying factor β. For instance, if $r_e = 6.5\ \Omega$ as in Example 7.4 and $\beta = 160$ (quite typical), the input impedance has increased to a level of

$$Z_i \cong \beta r_e = (160)(6.5\ \Omega) = \mathbf{1.04\ k\Omega}$$

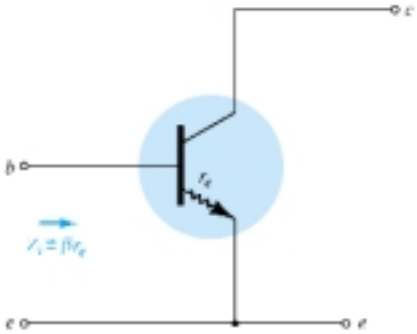


Figure 7.23 Impact of r_e on input impedance.

For the common-emitter configuration, typical values of Z_i defined by βr_e range from a few hundred ohms to the kilohm range, with maximums of about 6–7 kΩ.

For the output impedance, the characteristics of interest are the output set of Fig. 7.24. Note that the slope of the curves increases with increase in collector current. The steeper the slope, the less the level of output impedance (Z_o). The r_e model of Fig. 7.21 does not include an output impedance, but if available from a graphical analysis or from data sheets, it can be included as shown in Fig. 7.25.

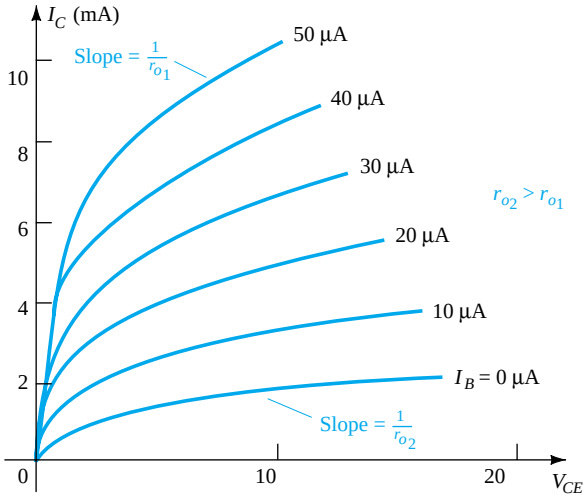


Figure 7.24 Defining r_o for the common-emitter configuration.

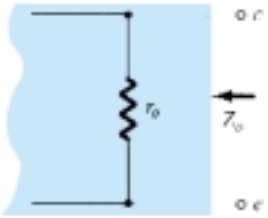


Figure 7.25 Including r_o in the transistor equivalent circuit.

For the common-emitter configuration, typical values of Z_o are in the range of 40 to 50 kΩ.

For the model of Fig. 7.25, if the applied signal is set to zero, the current I_c is 0 A and the output impedance is

$$\boxed{Z_o = r_o}_{CE} \tag{7.20}$$

Of course, if the contribution due to r_o is ignored as in the r_e model, the output impedance is defined by $Z_o = \infty\ \Omega$.

The voltage gain for the common-emitter configuration will now be determined for the configuration of Fig. 7.26 using the assumption that $Z_o = \infty\ \Omega$. The effect of including r_o will be considered in Chapter 8. For the defined direction of I_o and polarity of V_o,

$$V_o = -I_o R_L$$

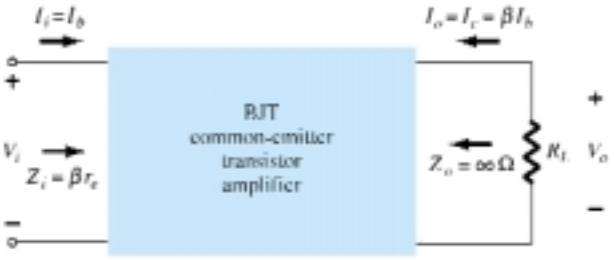


Figure 7.26 Determining the voltage and current gain for the common-emitter transistor amplifier.

The minus sign simply reflects the fact that the direction of I_o in Fig. 7.26 would establish a voltage V_o with the opposite polarity. Continuing gives

$$V_o = -I_o R_L = -I_c R_L = -\beta I_b R_L$$

and

$$V_i = I_i Z_i = I_b \beta r_e$$

so that

$$A_v = \frac{V_o}{V_i} = -\frac{\beta I_b R_L}{I_b \beta r_e}$$

and

$$\boxed{A_v = -\frac{R_L}{r_e}}_{CE, r_o = \infty\Omega} \tag{7.21}$$

The resulting minus sign for the voltage gain reveals that the output and input voltages are 180° out of phase.

The current gain for the configuration of Fig. 7.26:

$$A_i = \frac{I_o}{I_i} = \frac{I_c}{I_b} = \frac{\beta I_b}{I_b}$$

and

$$\boxed{A_i = \beta}_{CE, r_o = \infty\,\Omega} \tag{7.22}$$

Using the facts that the input impedance is βr_e, the collector current is βI_b, and the output impedance is r_o, the equivalent model of Fig. 7.27 can be an effective tool in the analysis to follow. For typical parameter values, the common-emitter configuration can be considered one that has a moderate level of input impedance, a high voltage and current gain, and an output impedance that may have to be included in the network analysis.

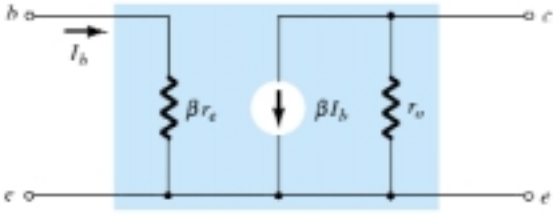


Figure 7.27 r_e model for the common-emitter transistor configuration.

EXAMPLE 7.5

Given $\beta = 120$ and $I_E = 3.2$ mA for a common-emitter configuration with $r_o = \infty\,\Omega$, determine:

(a) Z_i.

(b) A_v if a load of 2 kΩ is applied.

(c) A_i with the 2 kΩ load.

Solution

(a) $r_e = \dfrac{26\text{ mV}}{I_E} = \dfrac{26\text{ mV}}{3.2\text{ mA}} = 8.125\ \Omega$

and $Z_i = \beta r_e = (120)(8.125\ \Omega) = \mathbf{975\ \Omega}$

(b) Eq. (7.21): $A_v = -\dfrac{R_L}{r_e} = -\dfrac{2\text{ k}\Omega}{8.125\ \Omega} = \mathbf{-246.15}$

(c) $A_i = \dfrac{I_o}{I_i} = \beta = \mathbf{120}$

Common Collector Configuration

For the common-collector configuration, the model defined for the common-emitter configuration of Fig. 7.21 is normally applied rather than defining a model for the common-collector configuration. In subsequent chapters, a number of common-collector configurations will be investigated and the impact of using the same model will become quite apparent.

7.6 THE HYBRID EQUIVALENT MODEL

It was pointed out in Section 7.5 that the r_e model for a transistor is sensitive to the dc level of operation of the amplifier. The result is an input resistance that will vary with the dc operating point. For the hybrid equivalent model to be described in this section, the parameters are defined at an operating point that may or may not reflect the actual operating conditions of the amplifier. This is due to the fact that specification sheets cannot provide parameters for an equivalent circuit at every possible operating point. They must choose operating conditions that they believe reflect the general characteristics of the device. The hybrid parameters as shown in Fig. 7.28 are drawn from the specification sheet for the 2N4400 transistor described in Chapter 3. The values are provided at a dc collector current of 1 mA and a collector-to-emitter voltage of 10 V. In addition, a range of values is provided for each parameter for guidance in the initial design or analysis of a system. One obvious advantage of the specification sheet listing is the immediate knowledge of typical levels for the parameters of the device as compared to other transistors.

The quantities h_{ie}, h_{re}, h_{fe}, and h_{oe} of Fig. 7.28 are called the hybrid parameters and are the components of a small-signal equivalent circuit to be described shortly. For years, the hybrid model with all its parameters was the chosen model for the educational and industrial communities. Presently, however, the r_e model is applied more frequently, but often with the h_{oe} parameter of the hybrid equivalent model to provide

		Min.	Max.	
Input impedance (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz) 2N4400	h_{ie}	0.5	7.5	kΩ
Voltage feedback ratio (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{re}	0.1	8.0	$\times 10^{-4}$
Small-signal current gain (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz) 2N4400	h_{fe}	20	250	—
Output admittance (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{oe}	1.0	30	1μS

Figure 7.28 Hybrid parameters for the 2N4400 transistor.

some measure for the output impedance. Since specification sheets do provide the hybrid parameters and the hybrid model continues to receive a good measure of attention, it is quite important that the hybrid model be covered in some detail in this book. Once developed, the similarities between the r_e and hybrid models will be quite apparent. In fact, once the components of one are defined for a particular operating point, the parameters of the other model are immediately available.

Our description of the hybrid equivalent model will begin with the general two-port system of Fig. 7.29. The following set of equations (7.23) is only one of a number of ways in which the four variables of Fig. 7.29 can be related. It is the most frequently employed in transistor circuit analysis, however, and therefore is discussed in detail in this chapter.

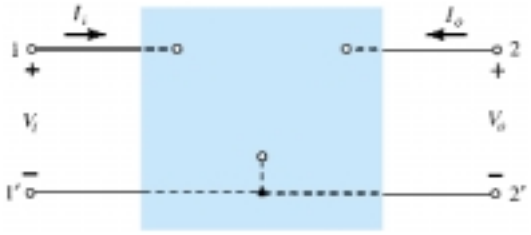


Figure 7.29 Two-port system.

$$V_i = h_{11}I_i + h_{12}V_o \tag{7.23a}$$

$$I_o = h_{21}I_i + h_{22}V_o \tag{7.23b}$$

The parameters relating the four variables are called *h-parameters* from the word "hybrid." The term *hybrid* was chosen because the mixture of variables (V and I) in each equation results in a "hybrid" set of units of measurement for the h-parameters. A clearer understanding of what the various h-parameters represent and how we can determine their magnitude can be developed by isolating each and examining the resulting relationship.

If we arbitrarily set $V_o = 0$ (short circuit the output terminals) and solve for h_{11} in Eq. (7.23a), the following will result:

$$h_{11} = \left.\frac{V_i}{I_i}\right|_{V_o = 0} \quad \text{ohms} \tag{7.24}$$

The ratio indicates that the parameter h_{11} is an impedance parameter with the units of ohms. Since it is the ratio of the *input* voltage to the *input* current with the output terminals *shorted*, it is called the *short-circuit input-impedance parameter.* The subscript 11 of h_{11} defines the fact that the parameter is determined by a ratio of quantities measured at the input terminals.

If I_i is set equal to zero by opening the input leads, the following will result for h_{12}:

$$h_{12} = \left.\frac{V_i}{V_o}\right|_{I_i = 0} \quad \text{unitless} \tag{7.25}$$

The parameter h_{12}, therefore, is the ratio of the input voltage to the output voltage with the input current equal to zero. It has no units since it is a ratio of voltage levels and is called the *open-circuit reverse transfer voltage ratio parameter.* The subscript 12 of h_{12} reveals that the parameter is a transfer quantity determined by a ratio of input to output measurements. The first integer of the subscript defines the

measured quantity to appear in the numerator; the second integer defines the source of the quantity to appear in the denominator. The term *reverse* is included because the ratio is an input voltage over an output voltage rather than the reverse ratio typically of interest.

If in Eq. (7.23b) V_o is equal to zero by again shorting the output terminals, the following will result for h_{21}:

$$h_{21} = \left.\frac{I_o}{I_i}\right|_{V_o=0} \qquad \text{unitless} \tag{7.26}$$

Note that we now have the ratio of an output quantity to an input quantity. The term *forward* will now be used rather than *reverse* as indicated for h_{12}. The parameter h_{21} is the ratio of the output current to the input current with the output terminals shorted. This parameter, like h_{12}, has no units since it is the ratio of current levels. It is formally called the *short-circuit forward transfer current ratio parameter.* The subscript 21 again indicates that it is a transfer parameter with the output quantity in the numerator and the input quantity in the denominator.

The last parameter, h_{22}, can be found by again opening the input leads to set $I_1 = 0$ and solving for h_{22} in Eq. (7.23b):

$$h_{22} = \left.\frac{I_o}{V_o}\right|_{I_i=0} \qquad \text{siemens} \tag{7.27}$$

Since it is the ratio of the output current to the output voltage, it is the output conductance parameter and is measured in siemens (S). It is called the *open-circuit output admittance parameter.* The subscript 22 reveals that it is determined by a ratio of output quantities.

Since each term of Eq. (7.23a) has the unit volt, let us apply Kirchhoff's voltage law "in reverse" to find a circuit that "fits" the equation. Performing this operation will result in the circuit of Fig. 7.30. Since the parameter h_{11} has the unit ohm, it is represented by a resistor in Fig. 7.30. The quantity h_{12} is dimensionless and therefore simply appears as a multiplying factor of the "feedback" term in the input circuit.

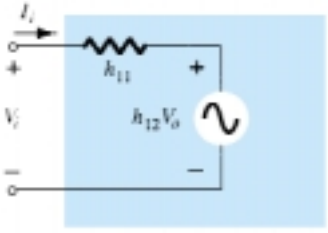


Figure 7.30 Hybrid input equivalent circuit.

Since each term of Eq. (7.23b) has the units of current, let us now apply Kirchhoff's current law "in reverse" to obtain the circuit of Fig. 7.31. Since h_{22} has the units of admittance, which for the transistor model is conductance, it is represented by the resistor symbol. Keep in mind, however, that the resistance in ohms of this resistor is equal to the reciprocal of conductance ($1/h_{22}$).

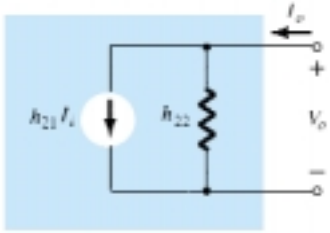


Figure 7.31 Hybrid output equivalent circuit.

The complete "ac" equivalent circuit for the basic three-terminal linear device is indicated in Fig. 7.32 with a new set of subscripts for the *h*-parameters. The notation of Fig. 7.32 is of a more practical nature since it relates the *h*-parameters to the resulting ratio obtained in the last few paragraphs. The choice of letters is obvious from the following listing:

$h_{11} \rightarrow$ *i*nput resistance $\rightarrow h_i$

$h_{12} \rightarrow$ *r*everse transfer voltage ratio $\rightarrow h_r$

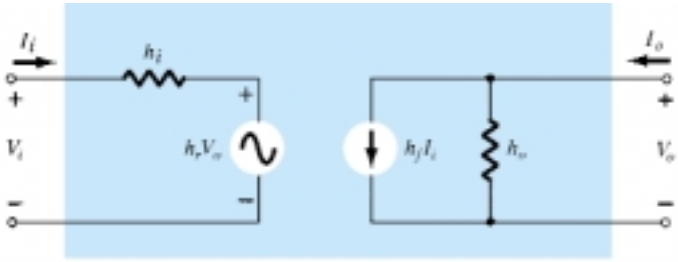


Figure 7.32 Complete hybrid equivalent circuit.

$h_{21} \rightarrow$ *f*orward transfer current ratio $\rightarrow h_f$

$h_{22} \rightarrow$ *o*utput conductance $\rightarrow h_o$

The circuit of Fig. 7.32 is applicable to any linear three-terminal electronic device or system with no internal independent sources. For the transistor, therefore, even though it has three basic configurations, *they are all three-terminal configurations*, so that the resulting equivalent circuit will have the same format as shown in Fig. 7.32. In each case, the bottom of the input and output sections of the network of Fig. 7.32 can be connected as shown in Fig. 7.33 since the potential level is the same. Essentially, therefore, the transistor model is a three-terminal two-port system. The *h*-parameters, however, will change with each configuration. To distinguish which parameter has been used or which is available, a second subscript has been added to the *h*-parameter notation. For the common-base configuration, the lowercase letter *b* was added, while for the common-emitter and common-collector configurations, the letters *e* and *c* were added, respectively. The hybrid equivalent network for the common-emitter configuration appears with the standard notation in Fig. 7.33. Note that $I_i = I_b$, $I_o = I_c$, and through an application of Kirchhoff's current law, $I_e = I_b + I_c$. The input voltage is now V_{be}, with the output voltage V_{ce}. For the common-base configuration of Fig. 7.34, $I_i = I_e$, $I_o = I_c$ with $V_{eb} = V_i$ and $V_{cb} = V_o$. The networks of Figs. 7.33 and 7.34 are applicable for *pnp* or *npn* transistors.

The fact that both a Thévenin and Norton circuit appear in the circuit of Fig. 7.32 was further impetus for calling the resultant circuit a *hybrid* equivalent circuit. Two additional transistor equivalent circuits, not to be discussed in this text, called the

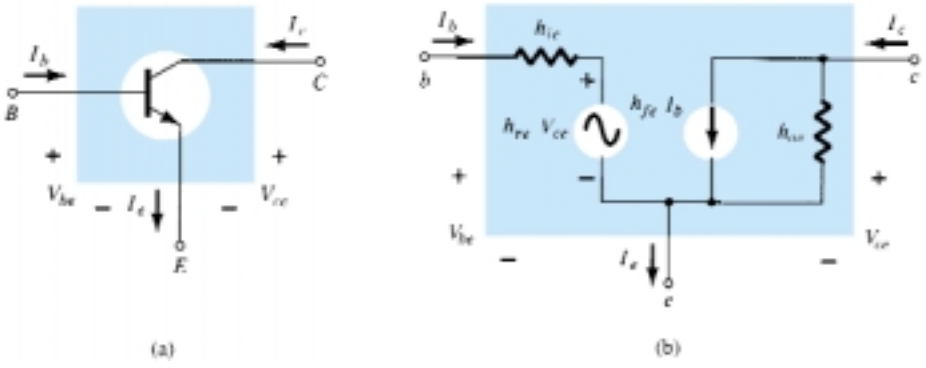


Figure 7.33 Common-emitter configuration: (a) graphical symbol; (b) hybrid equivalent circuit.

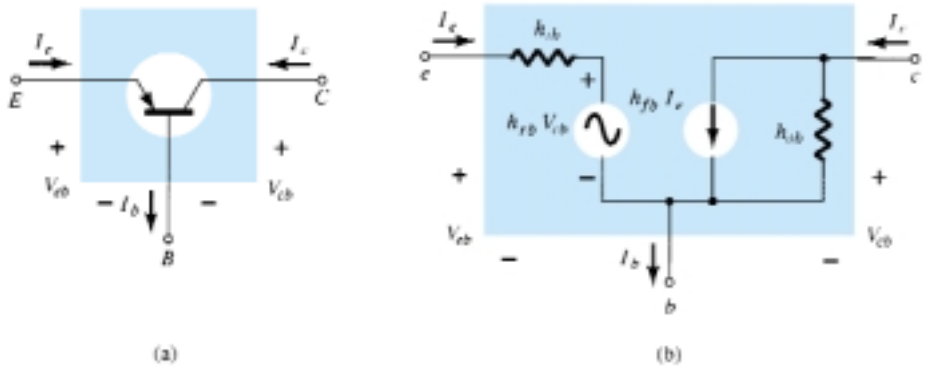


Figure 7.34 Common-base configuration: (a) graphical symbol; (b) hybrid equivalent circuit.

z-parameter and **y**-parameter equivalent circuits, use either the voltage source or the current source, but not both, in the same equivalent circuit. In Section 7.7, the magnitudes of the various parameters will be found from the transistor characteristics in the region of operation resulting in the desired *small-signal equivalent network* for the transistor.

For the common-emitter and common-base configurations, the magnitude of h_r and h_o is often such that the results obtained for the important parameters such as Z_i, Z_o, A_v, and A_i are only slightly affected if they (h_r and h_o) are not included in the model.

Since h_r is normally a relatively small quantity, its removal is approximated by $h_r \cong 0$ and $h_r V_o = 0$, resulting in a short-circuit equivalent for the feedback element as shown in Fig. 7.35. The resistance determined by $1/h_o$ is often large enough to be ignored in comparison to a parallel load, permitting its replacement by an open-circuit equivalent for the CE and CB models, as shown in Fig. 7.35.

The resulting equivalent of Fig. 7.36 is quite similar to the general structure of the common-base and common-emitter equivalent circuits obtained with the r_e model. In fact, the hybrid equivalent and the r_e models for each configuration have been repeated in Fig. 7.37 for comparison. It should be reasonably clear from Fig. 7.37a that

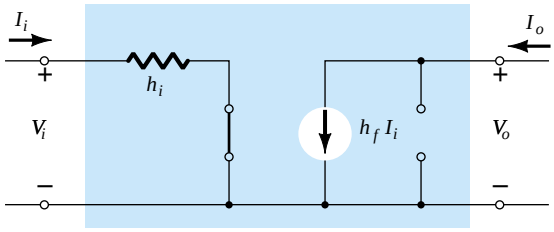


Figure 7.35 Effect of removing h_{re} and h_{oe} from the hybrid equivalent circuit.

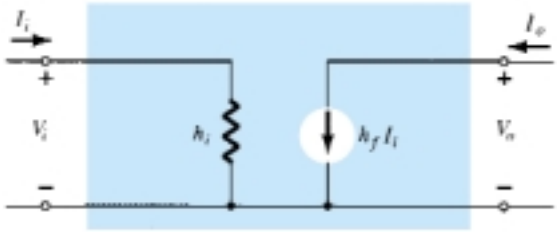


Figure 7.36 Approximate hybrid equivalent model.

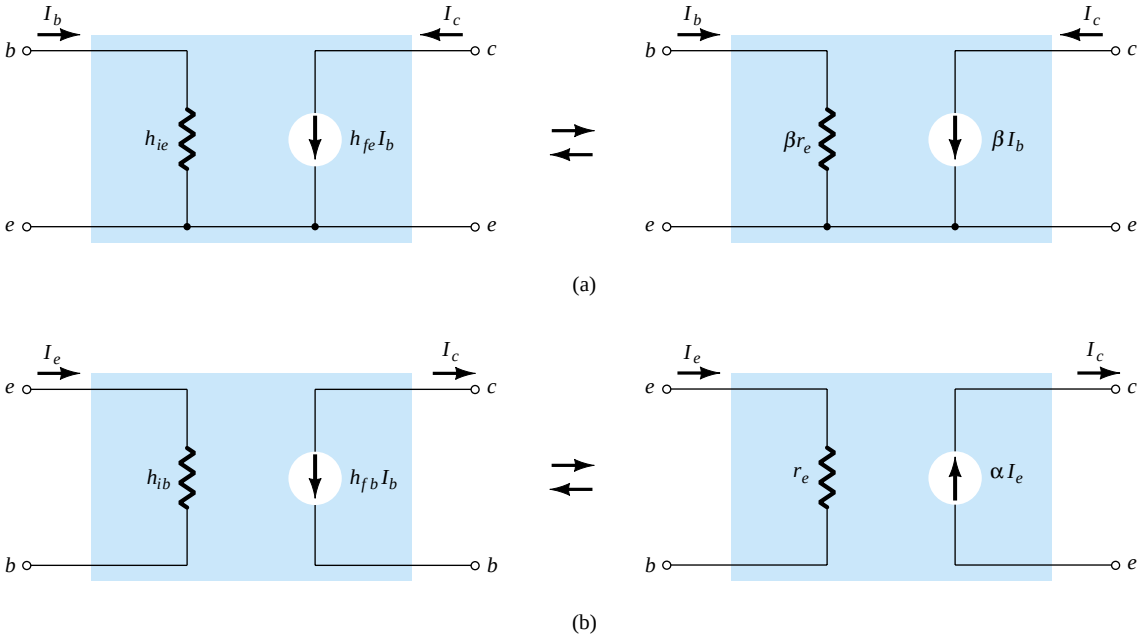


Figure 7.37 Hybrid versus r_e model: (a) common-emitter configuration; (b) common-base configuration.

$$h_{ie} = \beta r_e \tag{7.28}$$

and

$$h_{fe} = \beta_{ac} \tag{7.29}$$

From Fig. 7.37b,

$$h_{ib} = r_e \tag{7.30}$$

and

$$h_{fb} = -\alpha \cong -1 \tag{7.31}$$

In particular, note that the minus sign in Eq. (7.31) accounts for the fact that the current source of the standard hybrid equivalent circuit is pointing down rather than in the actual direction as shown in the r_e model of Fig. 7.37b.

EXAMPLE 7.6

Given $I_E = 2.5$ mA, $h_{fe} = 140$, $h_{oe} = 20$ μS (μmho), and $h_{ob} = 0.5$ μS, determine:
(a) The common-emitter hybrid equivalent circuit.
(b) The common-base r_e model.

Solution

(a) $$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.5 \text{ mA}} = 10.4\ \Omega$$

$$h_{ie} = \beta r_e = (140)(10.4\ \Omega) = \mathbf{1.456\ k\Omega}$$

$$r_o = \frac{1}{h_{oe}} = \frac{1}{20\ \mu\text{S}} = 50\ \text{k}\Omega$$

Note Fig. 7.38.

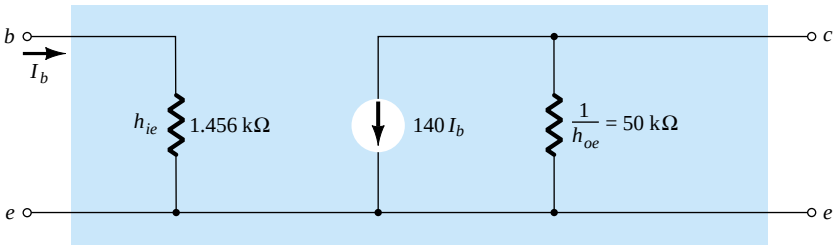


Figure 7.38 Common-emitter hybrid equivalent circuit for the parameters of example 7.6.

(b) $r_e = \mathbf{10.4\ \Omega}$

$$\alpha \cong 1, \qquad r_o = \frac{1}{h_{ob}} = \frac{1}{0.5\ \mu\text{S}} = \mathbf{2\ M\Omega}$$

Note Fig. 7.39.

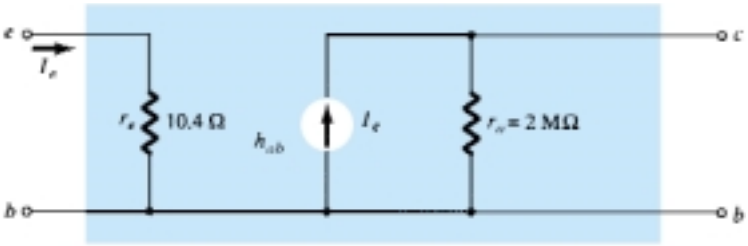


Figure 7.39 Common-base r_e model for the parameters of example 7.6.

A series of equations relating the parameters of each configuration for the hybrid equivalent circuit is provided in Appendix A. In Section 7.8, we demonstrate that the hybrid parameter h_{fe} (β_{ac}) is the least sensitive of the hybrid parameters to a change in collector current. Assuming, therefore, that $h_{fe} = \beta$ is a constant for the range of interest, is a fairly good approximation. It is $h_{ie} = \beta r_e$ that will vary significantly with I_C and should be determined at operating levels, since it can have a real impact on the gain levels of a transistor amplifier.

7.7 GRAPHICAL DETERMINATION OF THE *h*-PARAMETERS

Using partial derivatives (calculus), it can be shown that the magnitude of the *h*-parameters for the small-signal transistor equivalent circuit in the region of operation for the common-emitter configuration can be found using the following equations:*

$$h_{ie} = \frac{\partial v_i}{\partial i_i} = \frac{\partial v_{be}}{\partial i_b} \cong \left.\frac{\Delta v_{be}}{\Delta i_b}\right|_{V_{CE} = \text{ constant}} \quad \text{(ohms)} \tag{7.32}$$

$$h_{re} = \frac{\partial v_i}{\partial v_o} = \frac{\partial v_{be}}{\partial v_{ce}} \cong \left.\frac{\Delta v_{be}}{\Delta v_{ce}}\right|_{I_B = \text{ constant}} \quad \text{(unitless)} \tag{7.33}$$

$$h_{fe} = \frac{\partial i_o}{\partial i_i} = \frac{\partial i_c}{\partial i_b} \cong \left.\frac{\Delta i_c}{\Delta i_b}\right|_{V_{CE} = \text{ constant}} \quad \text{(unitless)} \tag{7.34}$$

$$h_{oe} = \frac{\partial i_o}{\partial v_o} = \frac{\partial i_c}{\partial v_{ce}} \cong \left.\frac{\Delta i_c}{\Delta v_{ce}}\right|_{I_B = \text{ constant}} \quad \text{(siemens)} \tag{7.35}$$

In each case, the symbol Δ refers to a small change in that quantity around the quiescent point of operation. In other words, the *h*-parameters are determined in the region of operation for the applied signal so that the equivalent circuit will be the most accurate available. The constant values of V_{CE} and I_B in each case refer to a condition that must be met when the various parameters are determined from the characteristics of the transistor. For the common-base and common-collector configurations, the proper equation can be obtained by simply substituting the proper values of v_i, v_o, i_i, and i_o.

The parameters h_{ie} and h_{re} are determined from the input or base characteristics, while the parameters h_{fe} and h_{oe} are obtained from the output or collector characteristics. Since h_{fe} is usually the parameter of greatest interest, we shall discuss the operations involved with equations, such as Eqs. (7.32) through (7.35), for this parameter first. The first step in determining any of the four hybrid parameters is to find the quiescent point of operation as indicated in Fig. 7.40. In Eq. (7.34) the condition $V_{CE} =$ constant requires that the changes in base current and collector current be taken along a vertical straight line drawn through the *Q*-point representing a fixed collector-to-emitter voltage. Equation (7.34) then requires that a small change in collector current be divided by the corresponding change in base current. For the greatest accuracy, these changes should be made as small as possible.

*The partial derivative $\partial v_i/\partial i_i$ provides a measure of the instantaneous change in v_i due to an instantaneous change in i_i.

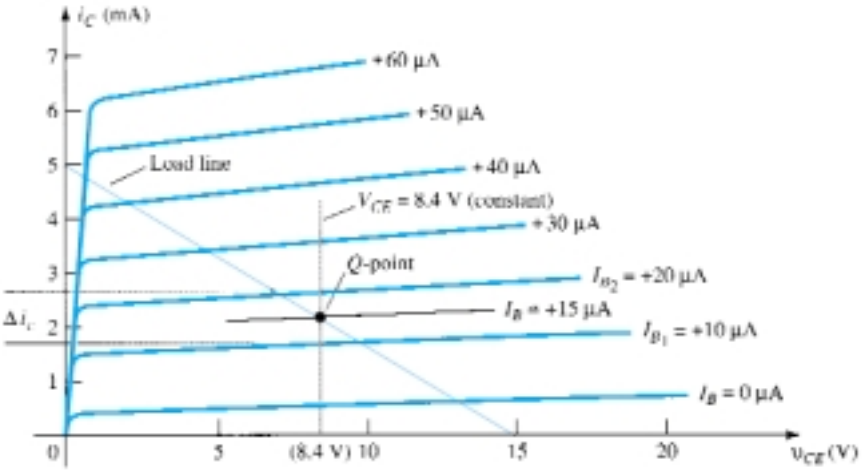


Figure 7.40 h_{fe} determination.

In Fig. 7.40, the change in i_b was chosen to extend from I_{B_1} to I_{B_2} along the perpendicular straight line at V_{CE}. The corresponding change in i_c is then found by drawing the horizontal lines from the intersections of I_{B_1} and I_{B_2} with V_{CE} = constant to the vertical axis. All that remains is to substitute the resultant changes of i_b and i_c into Eq. (7.34). That is,

$$|h_{fe}| = \left.\frac{\Delta i_c}{\Delta i_b}\right|_{V_{CE}\ =\ \text{constant}} = \left.\frac{(2.7 - 1.7)\ \text{mA}}{(20 - 10)\ \mu\text{A}}\right|_{V_{CE}\ =\ 8.4\ \text{V}}$$

$$= \frac{10^{-3}}{10 \times 10^{-6}} = \mathbf{100}$$

In Fig. 7.41, a straight line is drawn tangent to the curve I_B through the Q-point to establish a line I_B = constant as required by Eq. (7.35) for h_{oe}. A change in v_{CE} was then chosen and the corresponding change in i_C determined by drawing the horizontal lines to the vertical axis at the intersections on the I_B = constant line. Substituting into Eq. (7.35), we get

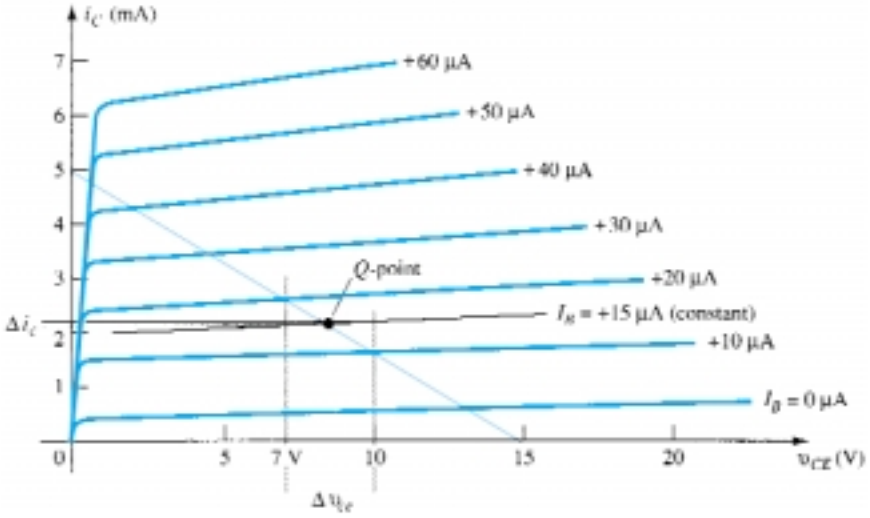


Figure 7.41 h_{oe} determination.

$$|h_{oe}| = \left.\frac{\Delta i_c}{\Delta v_{ce}}\right|_{I_B = \text{ constant}} = \left.\frac{(2.2 - 2.1)\text{ mA}}{(10 - 7)\text{ V}}\right|_{I_B = +15\ \mu\text{A}}$$

$$= \frac{0.1 \times 10^{-3}}{3} = \mathbf{33\ \mu A/V = 33 \times 10^{-6}\ S = 33\ \mu S}$$

To determine the parameters h_{ie} and h_{re} the Q-point must first be found on the input or base characteristics as indicated in Fig. 7.42. For h_{ie}, a line is drawn tangent to the curve $V_{CE} = 8.4$ V through the Q-point to establish a line $V_{CE} =$ constant as required by Eq. (7.32). A small change in v_{be} was then chosen, resulting in a corresponding change in i_b. Substituting into Eq. (7.32), we get

$$|h_{ie}| = \left.\frac{\Delta v_{be}}{\Delta i_b}\right|_{V_{CE} = \text{ constant}} = \left.\frac{(733 - 718)\text{ mV}}{(20 - 10)\ \mu\text{A}}\right|_{V_{CE} = 8.4\text{ V}}$$

$$= \frac{15 \times 10^{-3}}{10 \times 10^{-6}} = \mathbf{1.5\ k\Omega}$$

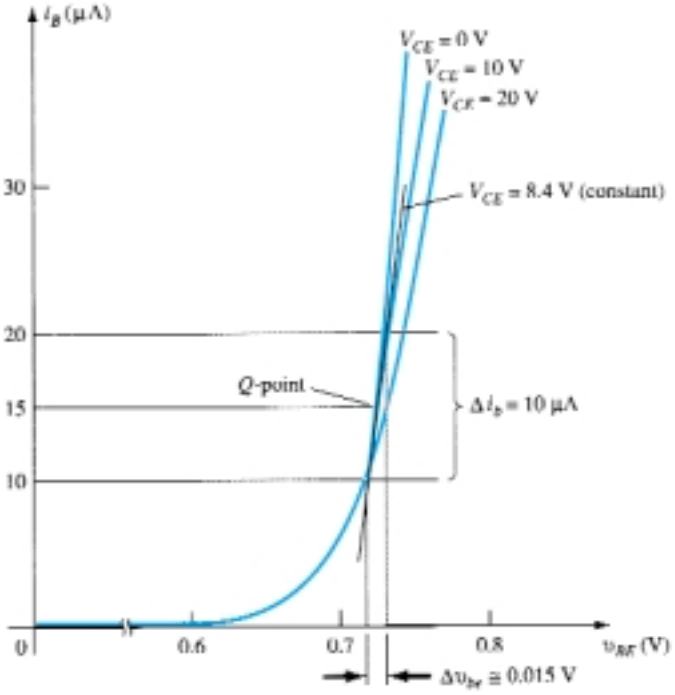


Figure 7.42 h_{ie} determination.

The last parameter, h_{re}, can be found by first drawing a horizontal line through the Q-point at $I_B = 15\ \mu$A. The natural choice then is to pick a change in v_{CE} and find the resulting change in v_{BE} as shown in Fig. 7.43.

Substituting into Eq. (7.33), we get

$$|h_{re}| = \left.\frac{\Delta v_{be}}{\Delta v_{ce}}\right|_{I_B = \text{ constant}} = \frac{(733 - 725)\text{ mV}}{(20 - 0)\text{ V}} = \frac{8 \times 10^{-3}}{20} = \mathbf{4 \times 10^{-4}}$$

For the transistor whose characteristics have appeared in Figs. 7.40 through 7.43, the resulting hybrid small-signal equivalent circuit is shown in Fig. 7.44.

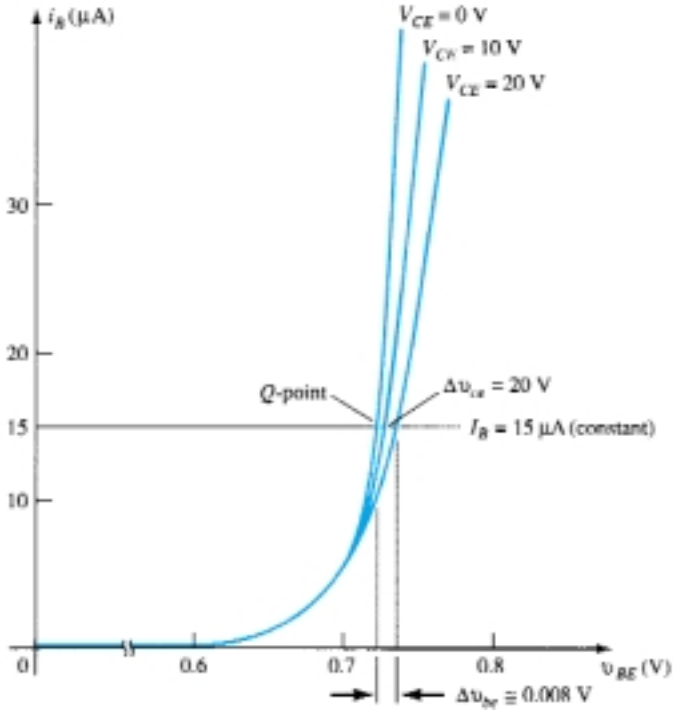


Figure 7.43 h_{re} determination.

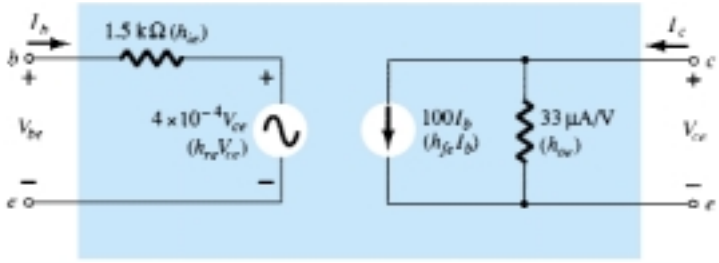


Figure 7.44 Complete hybrid equivalent circuit for a transistor having the characteristics that appear in Figs. 7.40 through 7.43.

As mentioned earlier, the hybrid parameters for the common-base and common-collector configurations can be found using the same basic equations with the proper variables and characteristics.

Table 7.1 lists typical parameter values in each of the three configurations for the broad range of transistors available today. The minus sign indicates that in Eq. (7.34) as one quantity increased in magnitude, within the change chosen, the other decreased in magnitude.

TABLE 7.1 Typical Parameter Values for the CE, CC, and CB Transistor Configurations

Parameter	*CE*	*CC*	*CB*
h_i	1 kΩ	1 kΩ	20 Ω
h_r	2.5×10^{-4}	≅1	3.0×10^{-4}
h_f	50	−50	−0.98
h_o	25 μA/V	25 μA/V	0.5 μA/V
$1/h_o$	40 kΩ	40 kΩ	2 MΩ

Note in retrospect (Section 3.5: Transistor Amplifying Action) that the input resistance of the common-base configuration is low, while the output resistance is high. Consider also that the short-circuit current gain is very close to 1. For the common-emitter and common-collector configurations, note that the input resistance is much higher than that of the common-base configuration and that the ratio of output to input resistance is about 40:1. Consider also for the common-emitter and common-base configurations that h_r is very small in magnitude. Transistors are available today with values of h_{fe} that vary from 20 to 600. For any transistor, the region of operation and conditions under which it is being used will have an effect on the various *h*-parameters. The effect of temperature and collector current and voltage on the *h*-parameters is discussed in Section 7.8.

7.8 VARIATIONS OF TRANSISTOR PARAMETERS

There are a large number of curves that can be drawn to show the variations of the *h*-parameters with temperature, frequency, voltage, and current. The most interesting and useful at this stage of the development include the *h*-parameter variations with junction temperature and collector voltage and current.

In Fig. 7.45, the effect of the collector current on the *h*-parameter has been indicated. Take careful note of the logarithmic scale on the vertical and horizontal axes. Logarithmic scales will be examined in Chapter 11. The parameters have all been normalized to unity so that the relative change in magnitude with collector current can easily be determined. On every set of curves, such as in Fig. 7.46, the operating point at which the parameters were found is always indicated. For this particular situation, the quiescent point is at the intersection of $V_{CE} = 5.0$ V and $I_C = 1.0$ mA. Since the frequency and temperature of operation will also affect the *h*-parameters, these quantities are also indicated on the curves. At 0.1 mA, h_{fe} is about 0.5 or 50% of its value at 1.0 mA, while at 3 mA, it is 1.5 or 150% of that value. In other words, if $h_{fe} = 50$ at $I_C = 1.0$ mA, h_{fe} has changed from a value of 0.5(50) = 25 to 1.5(50) = 75, with

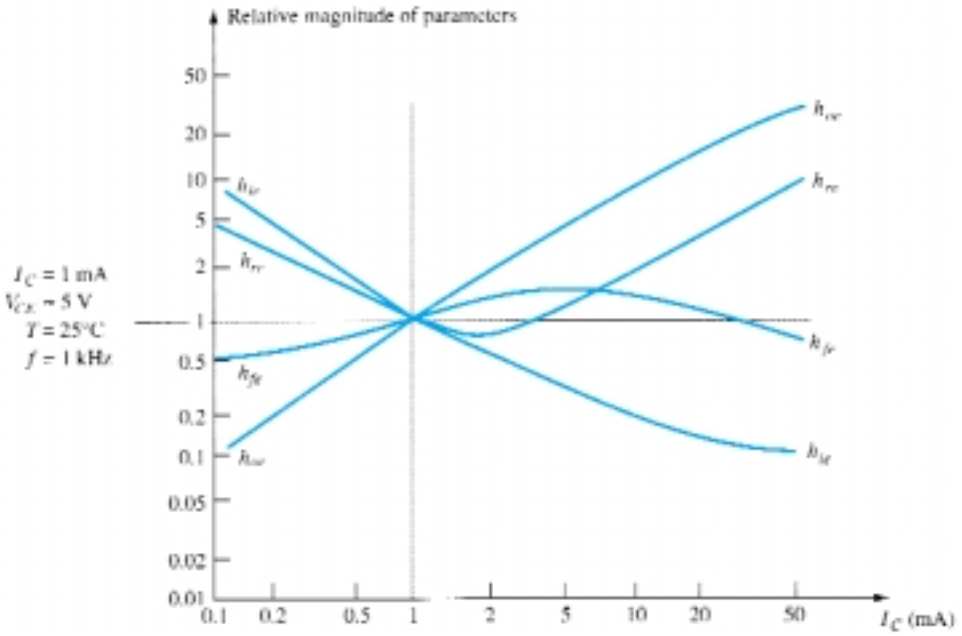


Figure 7.45 Hybrid parameter variations with collector current.

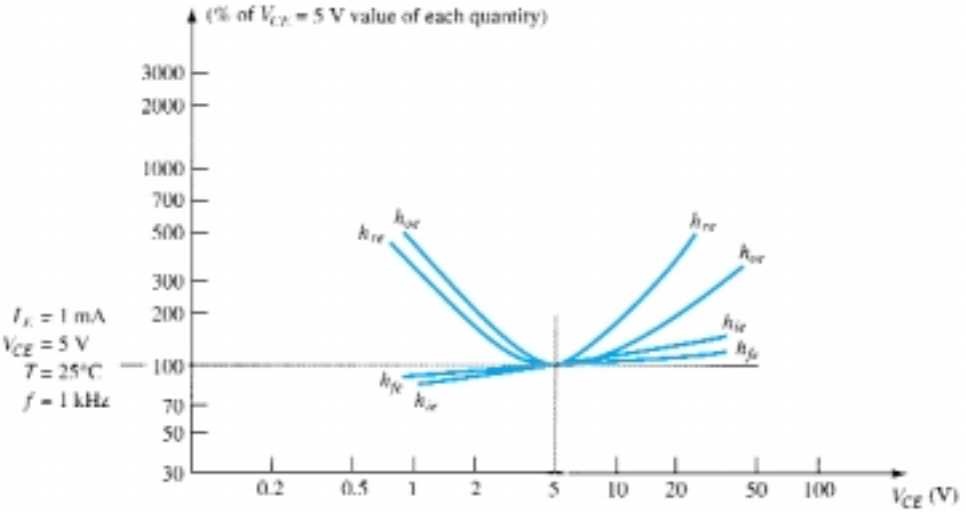


Figure 7.46 Hybrid parameter variations with collector–emitter potential.

a change of I_C from 0.1 to 3 mA. Consider, however, the point of operation at $I_C = 50$ mA. The magnitude of h_{re} is now approximately 11 times that at the defined Q-point, a magnitude that may not permit eliminating this parameter from the equivalent circuit. The parameter h_{oe} is approximately 35 times the normalized value. This increase in h_{oe} will decrease the magnitude of the output resistance of the transistor to a point where it may approach the magnitude of the load resistor. There would then be no justification in eliminating h_{oe} from the equivalent circuit on an approximate basis.

In Fig. 7.46, the variation in magnitude of the h-parameters on a normalized basis has been indicated with changes in collector voltage. This set of curves was normalized at the same operating point of the transistor discussed in Fig. 7.45 so that a comparison between the two sets of curves can be made. Note that h_{ie} and h_{fe} are relatively steady in magnitude while h_{oe} and h_{re} are much larger to the left and right of the chosen operating point. In other words, h_{oe} and h_{re} are much more sensitive to changes in collector voltage than are h_{ie} and h_{fe}.

It is interesting to note from Figs. 7.45 and 7.46 that the value of h_{fe} appears to change the least. Therefore, the specific value of current gain, whether h_{fe} or β, can, on an approximate and relative basis, be considered fairly constant for the range of collector current and voltage.

The value of $h_{ie} = \beta r_e$ does vary considerably with collector current as one might expect due to the sensitivity of r_e to emitter ($I_E \cong I_C$) current. It is therefore a quantity that should be determined as close to operating conditions as possible. For values below the specified V_{CE}, h_{re} is fairly constant, but it does increase measurably for higher values. It is indeed fortunate that for most applications the magnitude of h_{re} and h_{oe} are such that they can usually be ignored. They are quite sensitive to collector current and collector-to-emitter voltage.

In Fig. 7.47, the variation in h-parameters has been plotted for changes in junction temperature. The normalization value is taken to be room temperature: $T = 25°C$. The horizontal scale is a linear scale rather than a logarithmic scale as was employed for Figs. 7.45 and 7.46. In general, all the parameters increase in magnitude with temperature. The parameter least affected, however, is h_{oe}, while the input impedance h_{ie} changes at the greatest rate. The fact that h_{fe} will change from 50% of its normalized value at $-50°C$ to 150% of its normalized value at $+150°C$ indicates clearly that the operating temperature must be carefully considered in the design of transistor circuits.

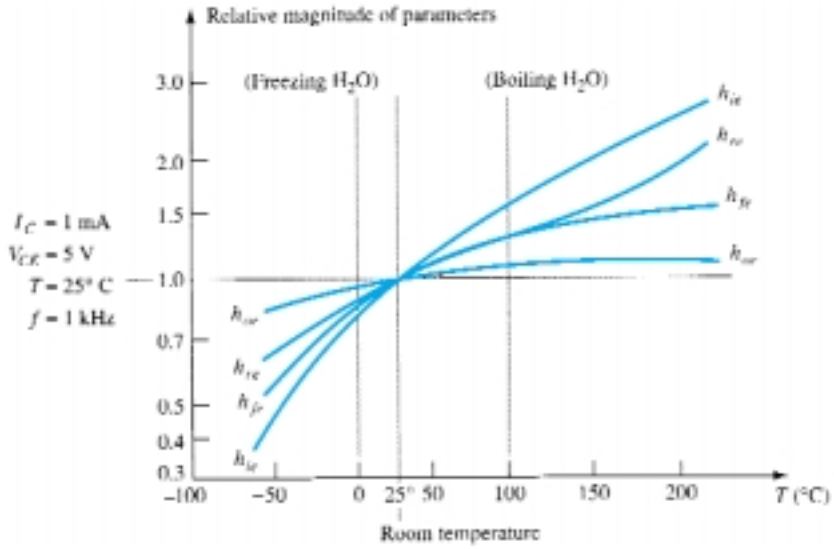


Figure 7.47 Hybrid parameter variations with temperature.

§ 7.2 Amplification in the AC Domain

PROBLEMS

1. (a) What is the expected amplification of a BJT transistor amplifier if the dc supply is set to zero volts?
(b) What will happen to the output ac signal if the dc level is insufficient? Sketch the effect on the waveform.
(c) What is the conversion efficiency of an amplifier in which the effective value of the current through a 2.2-kΩ load is 5 mA and the drain on the 18-V dc supply is 3.8 mA?

2. Can you think of an analogy that would explain the importance of the dc level on the resulting ac gain?

§ 7.3 BJT Transistor Modeling

3. What is the reactance of a 10-μF capacitor at a frequency of 1 kHz? For networks in which the resistor levels are typically in the kilohm range, is it a good assumption to use the short-circuit equivalence for the conditions just described? How about at 100 kHz?

4. Given the common-base configuration of Fig. 7.48, sketch the ac equivalent using the notation for the transistor model appearing in Fig. 7.5.

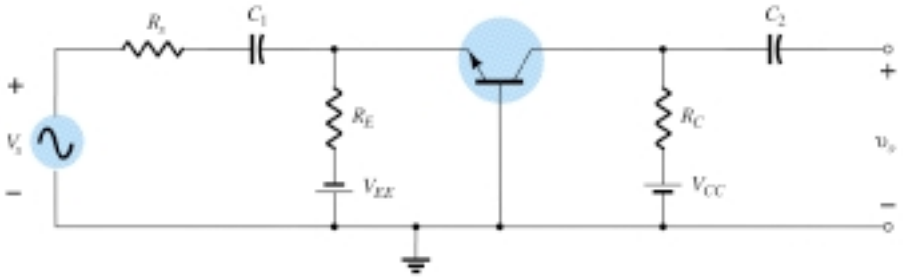


Figure 7.48 Problem 4

5. (a) Describe the differences between the r_e and hybrid equivalent models for a BJT transistor.
(b) For each model, list the conditions under which it should be applied.

§ 7.4 The Important Parameters: Z_i, Z_o, A_v, A_i

6. (a) For the configuration of Fig. 7.7, determine Z_i if $V_s = 40$ mV, $R_{\text{sense}} = 0.5$ kΩ, and $I_i = 20\ \mu$A.
(b) Using the results of part (a), determine V_i if the applied source is changed to 12 mV with an internal resistance of 0.4 kΩ.

7. (a) For the network of Fig. 7.10, determine Z_o if $V = 600$ mV, $R_{\text{sense}} = 10$ kΩ, and $I_o = 10\ \mu$A.
(b) Using the Z_o obtained in part (a), determine I_L for the configuration of Fig. 7.11 if $R_L = 2.2$ kΩ and $I_{\text{amplifier}} = 6$ mA.

8. Given the BJT configuration of Fig. 7.49, determine:
(a) V_i.
(b) Z_i.
(c) $A_{v_{\text{NL}}}$.
(d) A_{v_s}.

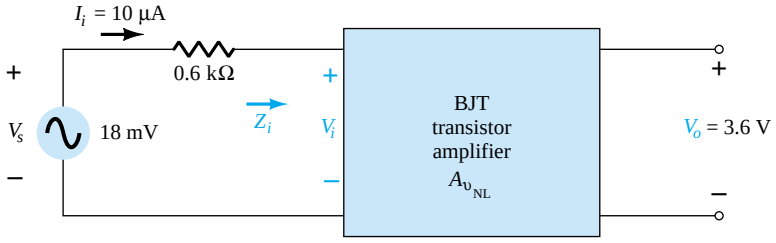


Figure 7.49 Problem 8

9. For the BJT amplifier of Fig. 7.50, determine:
(a) I_i.
(b) Z_i.
(c) V_o.
(d) I_o.
(e) A_i using the results of parts (a) and (d).
(f) A_i using Eq. (7.10).

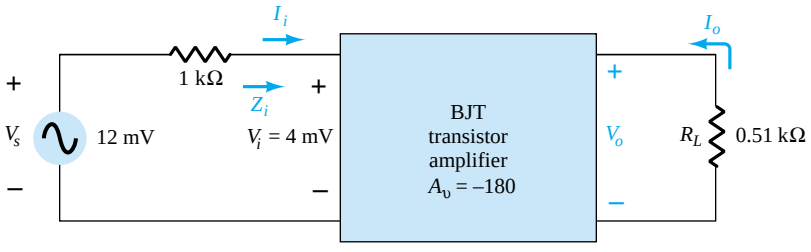


Figure 7.50 Problem 9

§ 7.5 The r_e Transistor Model

10. For the common-base configuration of Fig. 7.17, an ac signal of 10 mV is applied, resulting in an emitter current of 0.5 mA. If $\alpha = 0.980$, determine:
(a) Z_i.
(b) V_o if $R_L = 1.2$ kΩ.
(c) $A_v = V_o/V_i$.
(d) Z_o with $r_o = \infty$ Ω.
(e) $A_i = I_o/I_i$.
(f) I_b.

11. For the common-base configuration of Fig. 7.17, the emitter current is 3.2 mA and α is 0.99. Determine the following if the applied voltage is 48 mV and the load is 2.2 kΩ.
(a) r_e.
(b) Z_i.
(c) I_c.
(d) V_o.
(e) A_v.
(f) I_b.

12. Using the model of Fig. 7.27, determine the following for a common-emitter amplifier if $\beta = 80$, I_E (dc) = 2 mA, and $r_o = 40$ kΩ.
(a) Z_i.
(b) I_b.
(c) $A_i = I_o/I_i = I_L/I_b$ if $R_L = 1.2$ kΩ.
(d) A_v if $R_L = 1.2$ kΩ.

13. The input impedance to a common-emitter transistor amplifier is 1.2 kΩ with $\beta = 140$, $r_o = 50$ kΩ, and $R_L = 2.7$ kΩ. Determine:
(a) r_e.
(b) I_b if $V_i = 30$ mV.
(c) I_c.
(d) $A_i = I_o/I_i = I_L/I_b$.
(e) $A_v = V_o/V_i$.

§ 7.6 The Hybrid Equivalent Model

14. Given I_E (dc) = 1.2 mA, $\beta = 120$, and $r_o = 40$ kΩ, sketch the:
(a) Common-emitter hybrid equivalent model.
(b) Common-emitter r_e equivalent model.
(c) Common-base hybrid equivalent model.
(d) Common-base r_e equivalent model.

15. Given $h_{ie} = 2.4$ kΩ, $h_{fe} = 100$, $h_{re} = 4\times10^{-4}$, and $h_{oe} = 25$ μS, sketch the:
(a) Common-emitter hybrid equivalent model.
(b) Common-emitter r_e equivalent model.
(c) Common-base hybrid equivalent model.
(d) Common-base r_e equivalent model.

16. Redraw the common-emitter network of Fig. 7.3 for the ac response with the approximate hybrid equivalent model substituted between the appropriate terminals.

17. Redraw the network of Fig. 7.51 for the ac response with the r_e model inserted between the appropriate terminals. Include r_o.

18. Redraw the network of Fig. 7.52 for the ac response with the r_e model inserted between the appropriate terminals. Include r_o.

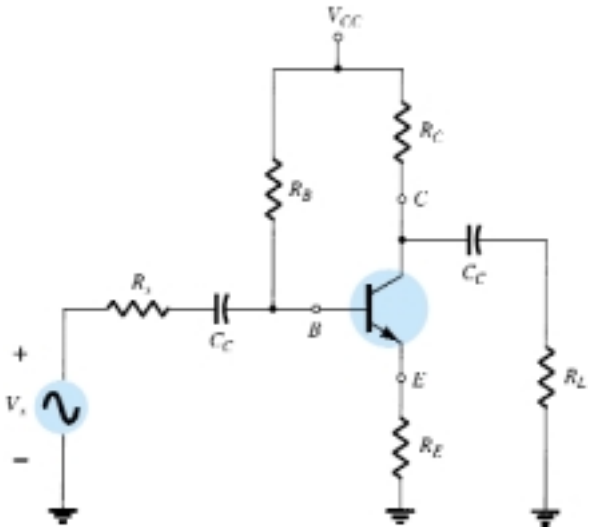


Figure 7.51 Problem 17

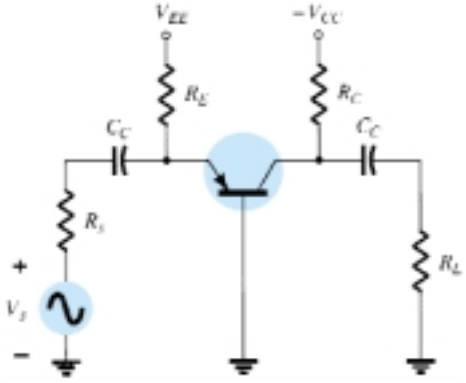


Figure 7.52 Problem 18

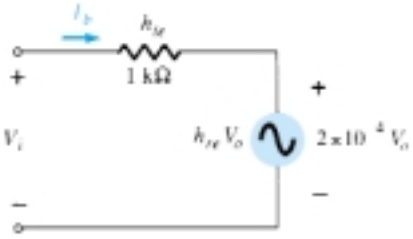


Figure 7.53 Problems 19, 21

19. Given the typical values of $h_{ie} = 1$ kΩ, $h_{re} = 2 \times 10^{-4}$, and $A_v = -160$ for the input configuration of Fig. 7.53:

(a) Determine V_o in terms of V_i.

(b) Calculate I_b in terms of V_i.

(c) Calculate I_b if $h_{re}V_o$ is ignored.

(d) Determine the percent difference in I_b using the following equation:

$$\%\ \text{difference in } I_b = \frac{I_b(\text{without } h_{re}) - I_b(\text{with } h_{re})}{I_b(\text{without } h_{re})} \times 100\%$$

(e) Is it a valid approach to ignore the effects of $h_{re}V_o$ for the typical values employed in this example?

20. Given the typical values of $R_L = 2.2$ kΩ and $h_{oe} = 20\ \mu$S, is it a good approximation to ignore the effects of $1/h_{oe}$ on the total load impedance? What is the percent difference in total loading on the transistor using the following equation?

$$\%\ \text{difference in total load} = \frac{R_L - R_L \| (1/h_{oe})}{R_L} \times 100\%$$

21. Repeat Problem 19 using the average values of the parameters of Fig. 7.28 with $A_v = -180$.

22. Repeat Problem 20 for $R_L = 3.3$ kΩ and the average value of h_{oe} in Fig. 7.28.

§ 7.7 Graphical Determination of the *h*-Parameters

23. (a) Using the characteristics of Fig. 7.40, determine h_{fe} at $I_C = 6$ mA and $V_{CE} = 5$ V.

(b) Repeat part (a) at $I_C = 1$ mA and $V_{CE} = 15$ V.

24. (a) Using the characteristics of Fig. 7.41, determine h_{oe} at $I_C = 6$ mA and $V_{CE} = 5$ V.
(b) Repeat part (a) at $I_C = 1$ mA and $V_{CE} = 15$ V.

25. (a) Using the characteristics of Fig. 7.42, determine h_{ie} at $I_B = 20\ \mu$A and $V_{CE} = 20$ V.
(b) Repeat part (a) at $I_B = 5\ \mu$A and $V_{CE} = 10$ V.

26. (a) Using the characteristics of Fig. 7.43, determine h_{re} at $I_B = 20\ \mu$A.
(b) Repeat part (a) at $I_B = 30\ \mu$A.

* **27.** Using the characteristics of Figs. 7.40 and 7.42, determine the approximate CE hybrid equivalent model at $I_B = 25\ \mu$A and $V_{CE} = 12.5$ V.

* **28.** Determine the CE r_e model at $I_B = 25\ \mu$A and $V_{CE} = 12.5$ V using the characteristics of Figs. 7.40 and 7.42.

* **29.** Using the results of Fig. 7.44, sketch the r_e equivalent model for the transistor having the characteristics appearing in Figs. 7.40 through 7.43. Include r_o.

§ 7.8 Variations of Transistor Parameters

For Problems 30 through 34, use Figs. 7.45 through 7.47.

30. (a) Using Fig. 7.45, determine the magnitude of the percent change in h_{fe} for an I_C change from 0.2 mA to 1 mA using the equation

$$\%\ \text{change} = \left| \frac{h_{fe}(0.2\ \text{mA}) - h_{fe}(1\ \text{mA})}{h_{fe}(0.2\ \text{mA})} \right| \times 100\%$$

(b) Repeat part (a) for an I_C change from 1 mA to 5 mA.

31. Repeat Problem 30 for h_{ie} (same changes in I_C).

32. (a) If $h_{oe} = 20\ \mu$S at $I_C = 1$ mA on Fig. 7.45, what is the approximate value of h_{oe} at $I_C = 0.2$ mA?
(b) Determine its resistive value at 0.2 mA and compare to a resistive load of 6.8 kΩ. Is it a good approximation to ignore the effects of $1/h_{oe}$ in this case?

33. (a) If $h_{oe} = 20\ \mu$S at $I_C = 1$ mA on Fig. 7.45, what is the approximate value of h_{oe} at $I_C = 10$ mA?
(b) Determine its resistive value at 10 mA and compare to a resistive load of 6.8 kΩ. Is it a good approximation to ignore the effects of $1/h_{oe}$ in this case?

34. (a) If $h_{re} = 2 \times 10^{-4}$ at $I_C = 1$ mA on Fig. 7.45, determine the approximate value of h_{re} at 0.1 mA.
(b) Using the value of h_{re} determined in part (a), can h_{re} be ignored as a good approximation if $A_v = 210$?

* **35.** (a) Reviewing the characteristics of Fig. 7.45, which parameter changed the least for the full range of collector current?
(b) Which parameter changed the most?
(c) What are the maximum and minimum values of $1/h_{oe}$? Is the approximation $1/h_{oe} \| R_L \cong R_L$ more appropriate at high or low levels of collector current?
(d) In which region of current spectrum is the approximation $h_{re}V_{ce} \cong 0$ the most appropriate?

36. (a) Reviewing the characteristics of Fig. 7.47, which parameter changed the most with increase in temperature?
(b) Which changed the least?
(c) What are the maximum and minimum values of h_{fe}? Is the change in magnitude significant? Was it expected?
(d) How does r_e vary with increase in temperature? Simply calculate its level at three or four points and compare their magnitudes.
(e) In which temperature range do the parameters change the least?

*Please Note: Asterisks indicate more difficult problems.